

Asset Pricing and Valuation

Lecture 8: Options & Greeks

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Options

Employee stock options

- What are employee stock options?
 - They are a **call** option on a company's stock granted by the company to its employees
 - Employee is able to take a stake in the company
 - If company does well, the share price increases above the **strike** price, employees gain by exercising their options
- Stock options are a way to attract and keep the best employees (offer attractive stock options)
- Microsoft - one of the first companies to offer stock options
- As Microsoft grew, estimated 10,000 employees became millionaires.
 - Now Microsoft only offers shares to employees

Employee stock options

- Popular with startup
 - Do not have resources to compete with top talent
 - Salaries supplemented with options
 - Company does well - enters into an IPO, the options are valuable.
- Employee options often last 10 - 15 years
- The **strike price** is often the **stock price** at the time of the grant (in-the-money)

Properties of employee stock options

- There is a vesting period where the option cannot be exercised
- When employees leave voluntarily or involuntarily during the vesting period, they forfeit their options
- When employees leave after the vesting period, they exercise their options at-the-money or in-the-money
- Not permitted to sell the options
- Upon option exercise, the company issues new shares and sells them to the employee for the **strike** price
- Early exercise
 - If employees want cash from the option - they must exercise their option - selling the underlying shares
 - i.e. cannot sell the option contract to somebody else.

Interests of shareholders and managers

- Do the interests of shareholders and managers align?
 - Interest of shareholders and managers need to be well aligned
 - Management motivated to make decisions in the best interests of shareholders (maximize shareholder wealth)
 - A way to motivate employees to work harder and longer hours
 - If company does well - everybody wins, management and employees
 - If company goes bankrupt - options are worthless (S_0 goes to zero and below K)
 - Employee stock options account for approximately 50% of remuneration of top executives in the US
- Executive stock options
 - If a company does well, shareholders value increases and executives are rewarded
 - Asymmetric payoff of options: If company does badly, shareholders lose wealth
 - Worst that happens to executives is that they do not make a gain
 - Better strategy is to give executives stock directly

Interests of shareholders and managers

- Suppose a manager wants to exercise their options in 3 months and sell their stock
- They may time announcements of **good news** and positive earnings to push up the stock price
- If they are given a new call contract “at-the-money” - so $S_0 = K$
 - Just before being given the contract they may take actions to reduce the stock price so the strike price will also be lower.
- Executive stock options may have the effect of motivating executives to focus on short term profits rather than long term growth
- Managers inside knowledge, able to affect outcomes and announcements may disadvantage shareholders

Greeks

- Each greek letter measures a different dimension to the risk in an option position
- The aim is to manage the Greeks so that all risks are acceptable

Delta intuition

- Δ in **calls** ranges from 0-1
 - Measures how much the price of an option changes if the stock changes by \$1
- Suppose we have a $\Delta = 0.6$
 - The options price goes up/down 60 cents for each dollar the asset (underlying) moves
- Suppose we have a $\Delta = 1$
 - The price of the option moves a dollar for each dollar the asset (underlying) moves
- Δ in **puts** ranges from -1 to 0

Delta intuition

- Assume the stock trades at 174.12
- We buy at the money call option at 175 strike for 0.79 cents
- The Δ is 0.468
- For a **call**
 - Every dollar the stock moves the options price moves 47 cents
 - If our stock went from 174.12 to 175.12 the option moves by 47 cents
 - So our new option contract is worth \$1.26 (0.79 + 0.47)
- For a **put**
 - A 175 strike put, we have a Δ of 0.352
 - If the stock moves down a dollar, the option will increase in price

Greeks

- Suppose we have sold a \$300, 000 European call option on 100,000 shares of a non-dividend-paying stock
- Assume the stock price is \$49 and strike price is \$50
- The risk free rate is 5% per annum, the stock price volatility is 20% per annum
- The time to maturity is 20 weeks (0.3846 years)
- The expected return from the stock is 13%

$$S_0 = 49, K = 50, r = 0.05, \sigma = 0.2, T = 0.3846, \mu = 0.13$$

- Suppose A Binomial Options Pricing model or a Black Scholes Model has priced the option about \$240, 000 (\$2.4 for an option to buy one share)
- We have therefore sold the **call** option for \$60, 00 more than its theoretical value (price we paid - price from Binomial/Black Scholes model : 300 - 240)
- We are faced with the **problem of hedging the risks.**

Naked and covered positions

- One strategy is to do nothing (taking a naked position)
- This works well if the stock price is less than \$50 in 20 weeks time
- The option then costs us nothing (expires worthless) and makes us a profit of \$300,000
- When the stock is greater than \$50 then the option is exercised
 - We are forced to buy 100,000 shares at the market price at $T = 20weeks$ to cover the call
 - Our cost is $100,000 \times (S_T - K)$
 - i.e. if the stock price is \$60, the option costs is \$1,000,000 i.e. $(100,000 * (60 - 50))$
 - Significantly higher than the \$300,000 charged for the option

Naked and covered positions

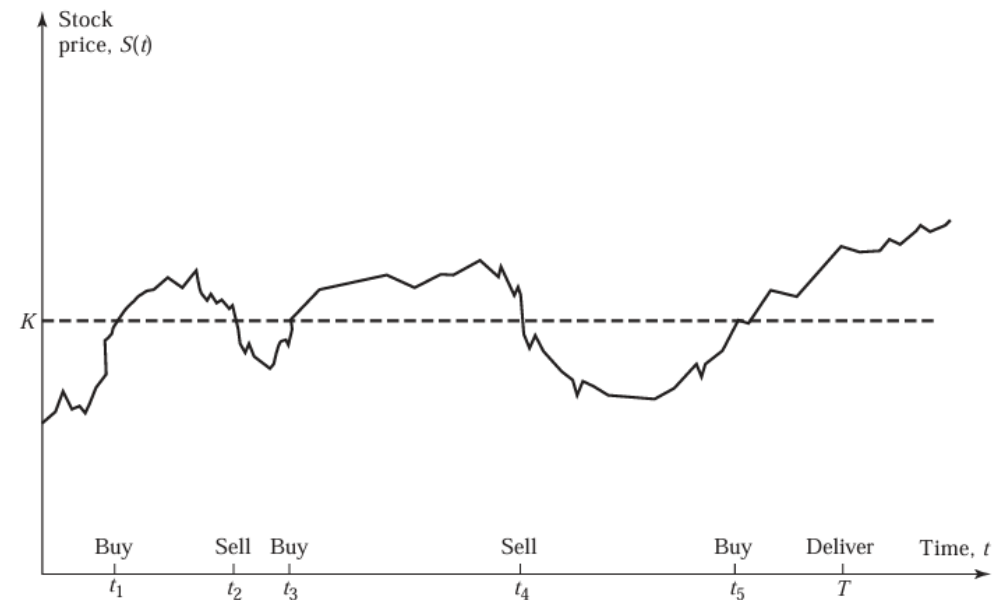
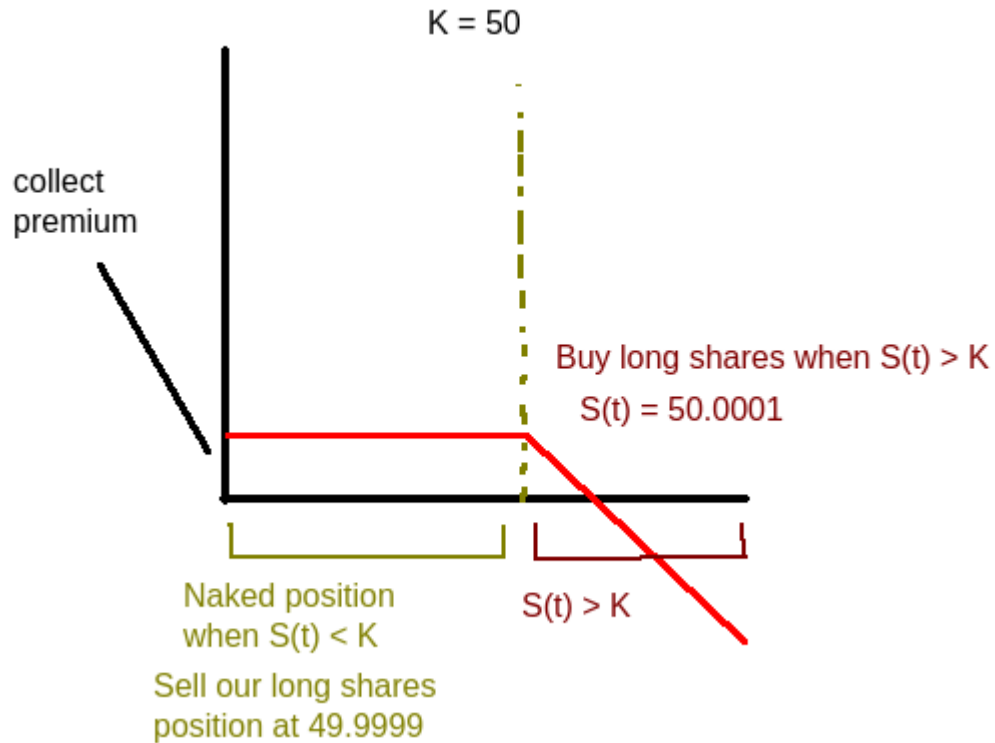
- Alternatively, we could take a **covered position**
- We buy 100,000 shares as soon as the option has been sold $t = 0$
- If the option is exercised, the strategy works well, otherwise it could result in a significant loss.
 - i.e. if stock price falls to \$40, we lose \$900,000 on the stock position (greater than \$300,000)
 - i.e. we buy a long position at 49 a share - stocks go down to 40 - losing 900,000
- So, neither a **naked** position or a **covered call** provides a good hedge
- A good hedge ensures that the cost is always close to \$240,000

Stop loss strategy

- Suppose we have written a call option with strike K for 1 unit of stock
- We buy the stock as soon as the price rises above K and sell it as soon as it falls below K
- Hold a naked position when $S_t < K$ and a covered position when $S_t > K$
 - So we own the stock if the option closes in the money
 - Do not own the stock if the option closes out of the money
- Stop loss strategy works well, but does not work as a **hedging** procedure.
- If stock price never reached the strike price K the **hedging** procedure costs nothing
- If the path of the stock price crosses the strike price many times, the procedure is expensive.

Stop loss strategy

Stop-loss covered call strategy

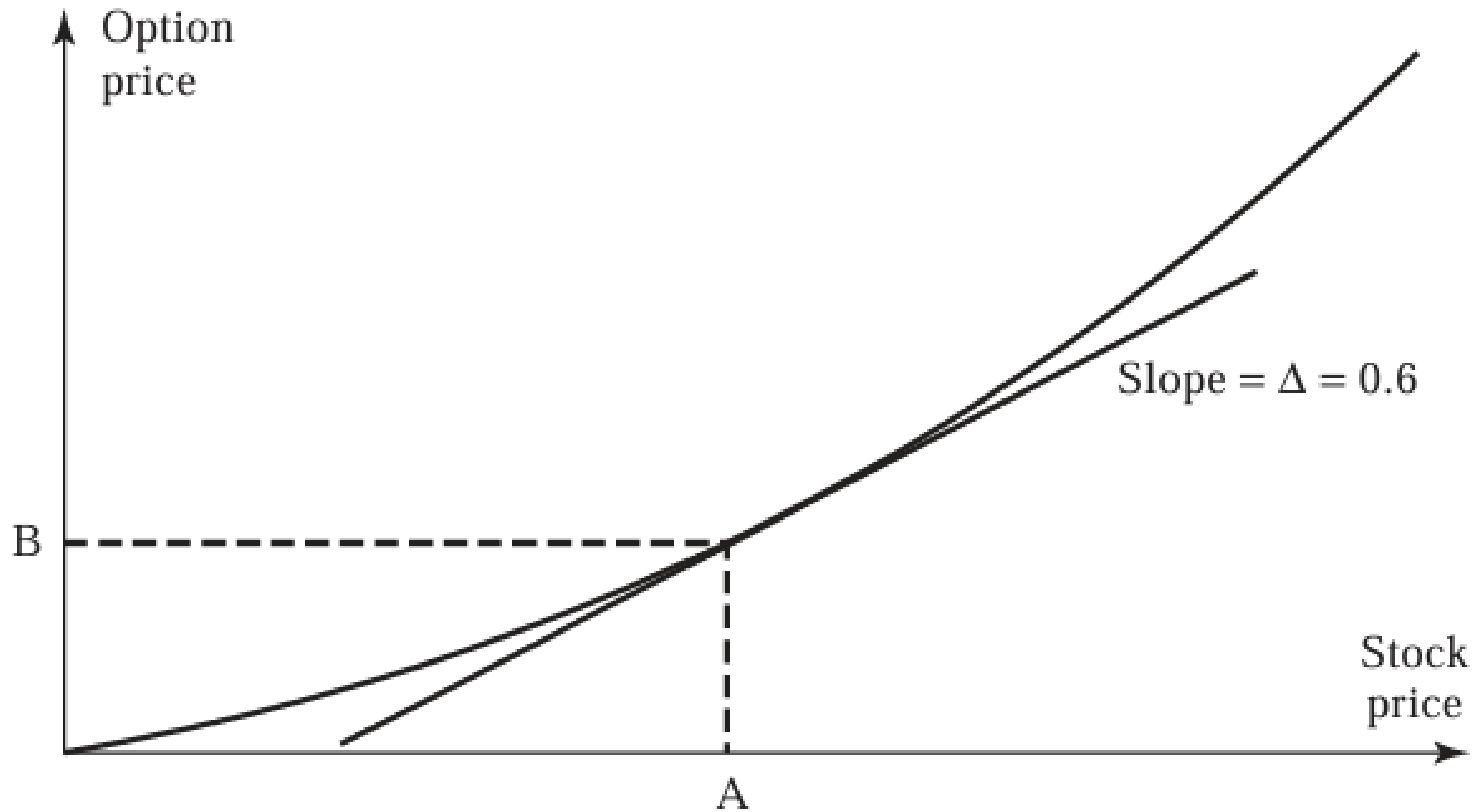


- I.e. we buy the stock at t_1 , sell it at t_2 , buy it again at t_3 , sell it at t_4 , buying it finally at t_5 and delivering it at T
- We purchase the stock at $K + \epsilon$ and sell the stock at $K - \epsilon$
- Every purchase and sale involves a cost
- We could observe stock prices more closely and try to reduce ϵ
- But as ϵ gets smaller, we make trades more frequently (transaction costs)
- As $\epsilon \rightarrow 0$, the number of trades tends to infinity

Delta hedging

- Δ is the rate of change of the option price with respect to the price of the underlying asset
- It is the slope of the curve that relates options prices to the underlying asset.
- Consider $\Delta = 0.6$
 - When a stock price changes by a small amount, the option price changes by 60% of that amount
 - Say a stock increased a \$1 then the options price increases by 0.60 cents.

Delta hedging



$$\Delta = \frac{f_u - f_d}{S_0u - S_0d}$$

Delta hedging

- When the stock corresponds to point A, the options price is priced at B
- Δ is the slope of the line indicated.

$$\Delta = \partial c / \partial S$$

- Where c is the price of the call option and S is the stock price.
- Suppose the stock price is \$100 and the option price is \$10
- If we sold 20 call option contract, (options on 2000 shares)
 - Assume that $\Delta = 0.6$
 - The investors position is hedged by buying $0.6 \times 2000 = 1,200$
 - The **gain** on the stock position will offset the **loss** on the option position
 - or, the **loss** on the stock position would offset the **gain** on the option position
- If stock price goes up by \$1 (gain of \$1200 on shares purchased) - the option price goes up $0.6 \times \$1$ (producing a loss of \$1200 of the options written)
- If stock prices goes down by \$1 (produces loss of \$1200 on shares purchased) - the option price will go down by $0.6 \times \$1$ (producing a gain of \$1200 on the options written)
- So Δ of our short position in 2000 options is;

$$0.6 \times (-2000) = -1200$$

- We lose $1200\Delta \times S$ on the option position when the stock price increases by ΔS
- Here $\Delta = 0.6$ but a position when the Δ is 0 is called **delta neutral**

Delta

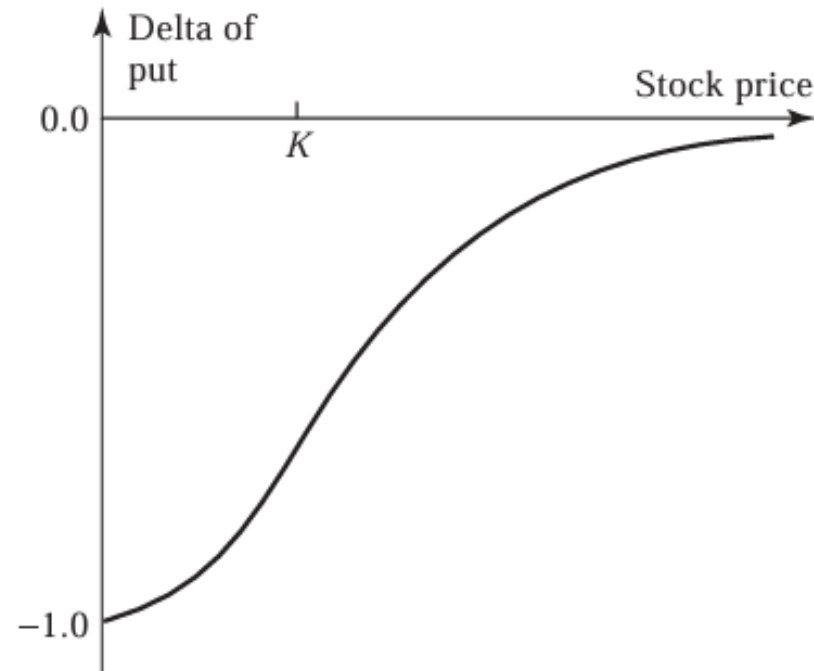
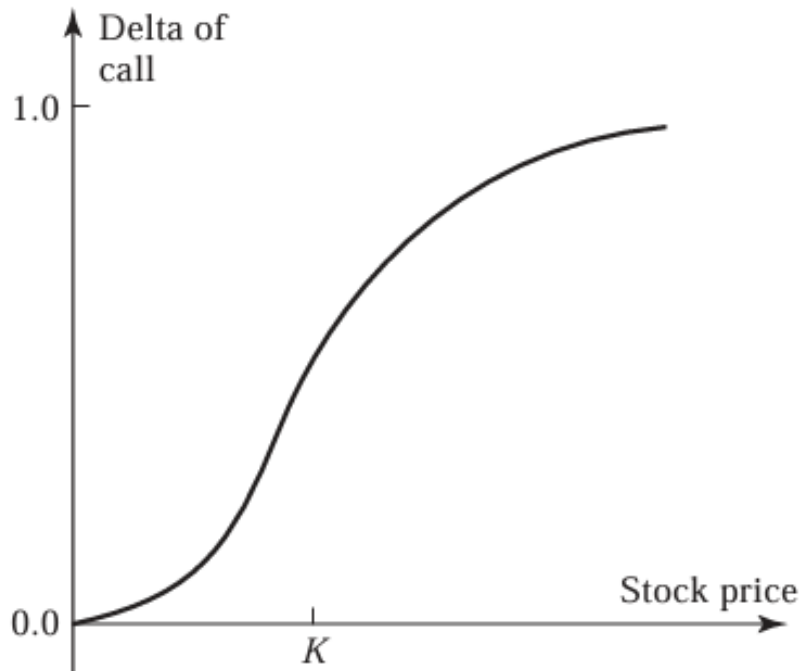
- In the previous example we stated that $\Delta = 0.6$ and is fixed
- In the real world the Δ is not constant
- If we have a Δ neutral hedge we need to adjust our positions often *rebalancing*
- i.e. If the stock price increased to \$10 then Δ would also increase (previous plot)
- If Δ rises from 0.60 to 0.65
 - We need to rebalance by $0.05 * 2000 = 100$ shares (buy 100 more shares)
 - Ensures we maintain the hedge *dynamic hedging* as opposed to *static hedging*
 - **static hedging** where a hedge is set up initially and never changed
 - **Dynamic hedge** - constantly changing with changing values of Δ

Delta

- We can set up a riskless portfolio consisting of a position in an option on a stock and a position in the stock itself
 - Expressed in terms of Δ
 - -1 : option
 - $+\Delta$ shares of stock
 - Options can be valued by setting up a Δ -neutral position

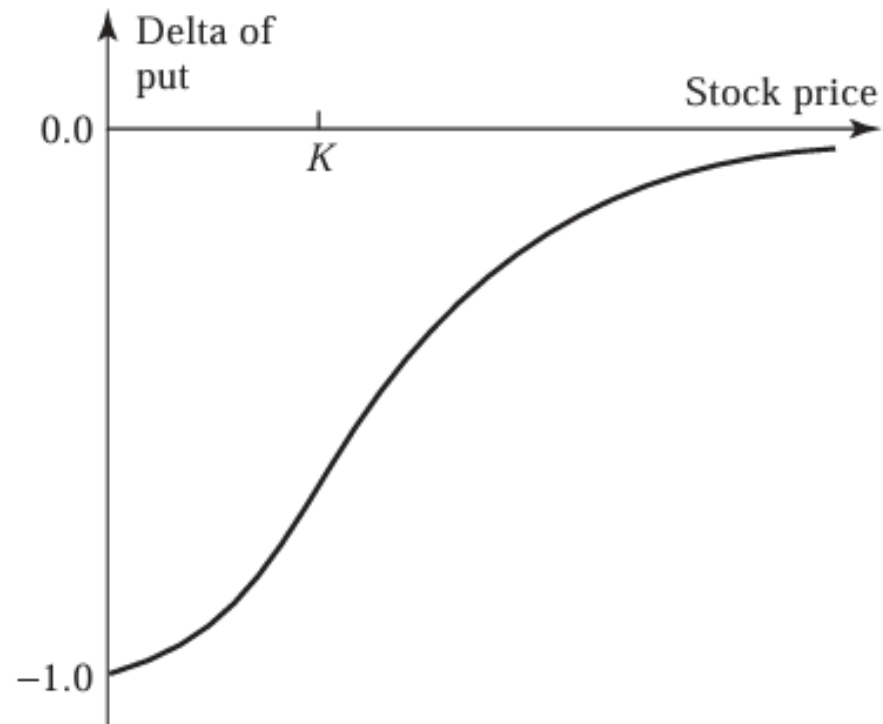
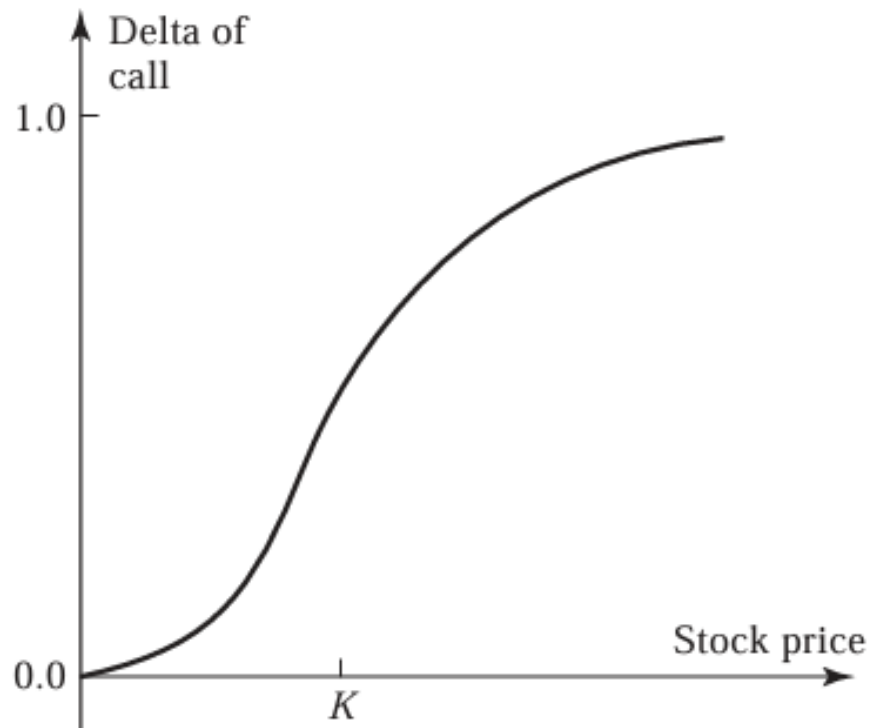
Delta

- Variation of the Δ **call** and **put** with changes in **stock price**
- **Call option**
- As stock prices increase, the delta of the **call** option increases
- When Δ is 0.5 then $S_t = K$ (at the money)
 - When $S_t < K$ (deep out the money call) the Δ is close to 0
 - suggesting that the call options price is not sensitive to changes in the stock price
 - When $S_t > K$ (deep in the money call) the Δ is close to 1
 - suggesting that the call options price is highly sensitive to changes in the stock price
 - Moves 1-for-1 with the stock price. - If stock price moves \$1 then the options price moves \$1



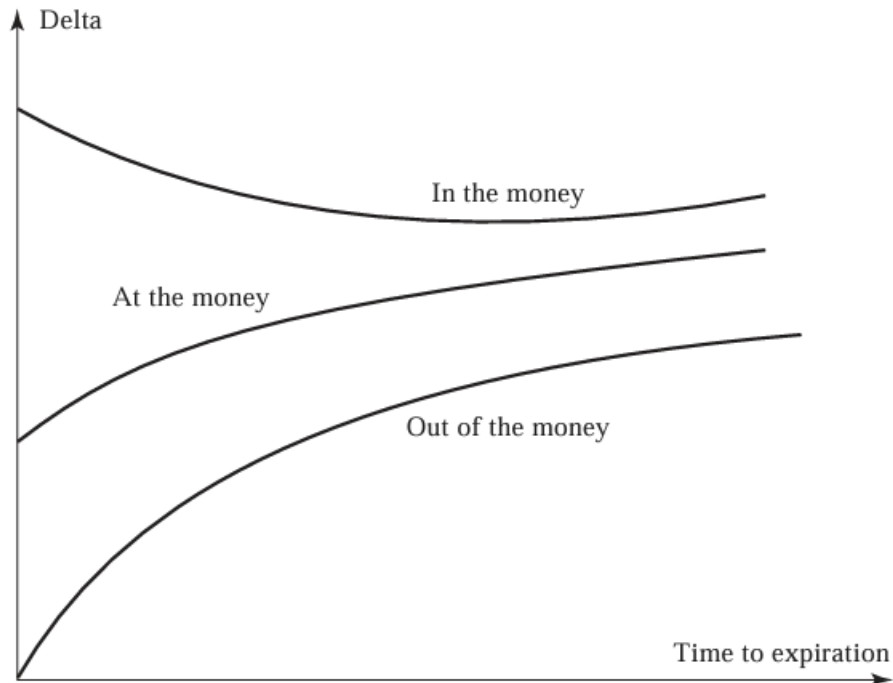
Delta

- Variation of the Δ **call** and **put** with changes in **stock price**
- **Put option**
- As stock price increases the Δ becomes less negative (close to zero)
- When Δ is -0.5 then $S_t = K$ (at the money)
 - When $S_t > K$ (deep out the money) Δ approaches 0
 - Put option price not sensitive to the stock price movements
 - When $S_t < K$ (deep in the money) Δ approaches -1
 - Suggesting the put options price changes almost one for one with the stock price (but in the opposite direction)



Delta

- Call option
- Variation of Δ with **time** for an in-the-money, at-the-money and out-of-the-money **call** option
- **In the money** - have a higher Δ ($S_t > K$) - price of the option is more sensitive to changes in the stock price
- As time decreases the Δ of in-the-money call options increases
- **At the money** - $S_t = K$ - Δ is close to 0.5 - suggests that options price changes at half the rate of the stock price
 - As time to expiry increases Δ increases because the option can be in the money or out the money
- **Out the money** - $S_t < K$ - Δ is lowest so movements in the stock price do not affect the options price much as we approach expiry



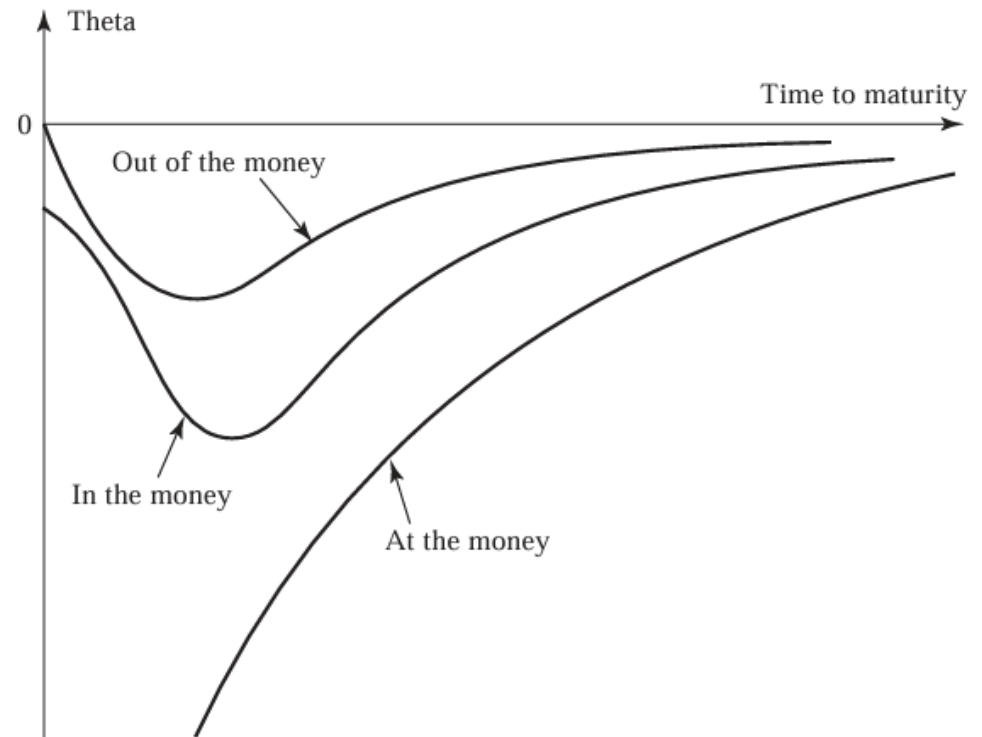
Theta (time decay)

- Θ of an option measures the rate of change of the value of the option with respect to the passage of time
- Referred to *time decay*
- When Θ is quoted its measured in days
 - So Θ is the change in the contracts value when 1 day passes (all else remaining the same)
 - Θ can be measures in *per calendar day* (365) or *per trading day* (252)

Theta (time decay)

Theta

- Θ measures the rate of decline in the options value with each passing day (due to time decay)
- **out of the money** ($S_t < K$) - As the option approaches maturity, the time value Θ decreases
- So the probability of it becoming in the money (profitable to exercise) decreases as we get closer to expiry.
- **At the money** ($S_t = K$) - Θ decreases more rapidly as the option approaches maturity
 - Since, the uncertainty about whether the option will will end up **in the money** or **out the money** is highest when the option is **at the money**
 - This uncertainty is rapidly falling as the option approaches expiration
- **In the money** - ($S_t > K$) - Θ decreases at a slower rate as we reach maturity compared with an **at-the money** option
 - Because there is less uncertainty about the option finishing in the money - but still it loses time value as the expiration approaches.



- For all options \ **Thata** approaches zero - since there is less time for the stock to move and less time value embedded in the options price
- non-linear time decay
 - Accelerating as time gets closer (characteristic of time decay)

Gamma

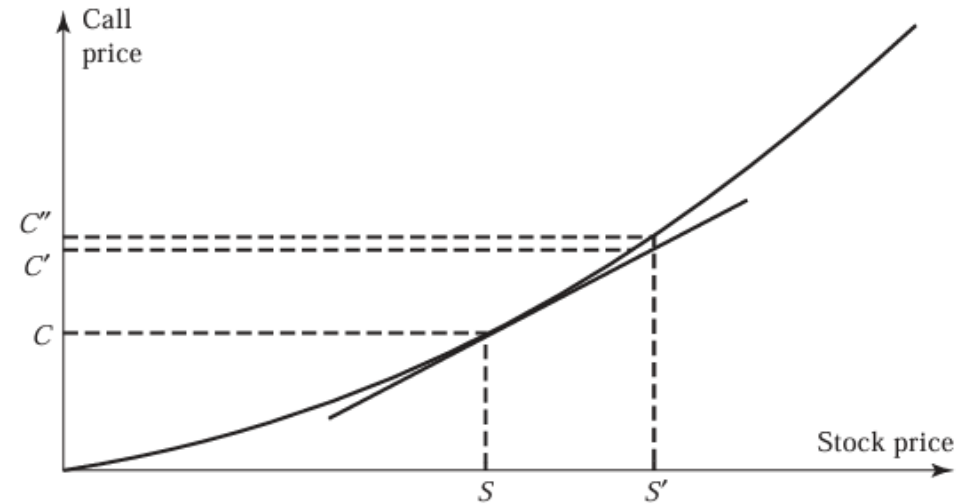
Γ measures the rate of change of delta with respect to changes in the underlying price +
Second partial derivative of the options price with respect to the underlying price

$$\Gamma = \frac{\partial^2 V}{\partial S^2}$$

- If Γ is small, then Δ changes slowly, and adjustments to keep a portfolio delta neutral need to be made only relatively infrequently.
- If Γ is highly positive/negative - the Δ \$ is very sensitive to the price of the underlying asset
 - Risky to leave a Δ neutral portfolio unchanged for a long period of time.
- Recall, the first derivative of the options price with respect to the underlying price is the delta
 - The second derivative is Γ
- Therefore, Γ measures the rate of change of Δ for a one-unit change in the stock price.

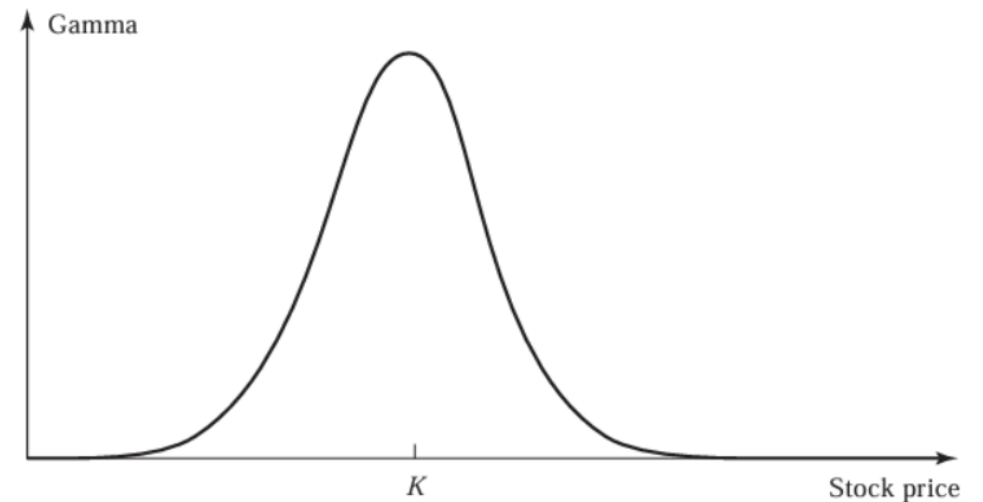
Gamma

- When the stock price increases from S to S' the Δ of the option (slope of the tangent line) also increases
 - The steepness indicates a high Γ
 - High Γ suggests that Δ is sensitive to changes in the stock price
- Γ is important for **at-the-money** options where Δ can change rapidly when the stock price moves around the strike price.
- A positive Γ means as the S_t increases, Δ also increases - leading to more than proportional increase in the call price.
- A negative Γ indicates that Δ decreases - as the stock price increases



Gamma

- Γ measures the rate of change of an options Δ per one-point move in the underlying stock price.
- Highest when we are **at-the-money** (stock price = strike)
- Increases and decreases when we are **out the money** or **in the money**
- High Γ means the options sensitivity to stock price changes is high - leads to larger changes in the options price as the stock price fluctuates.



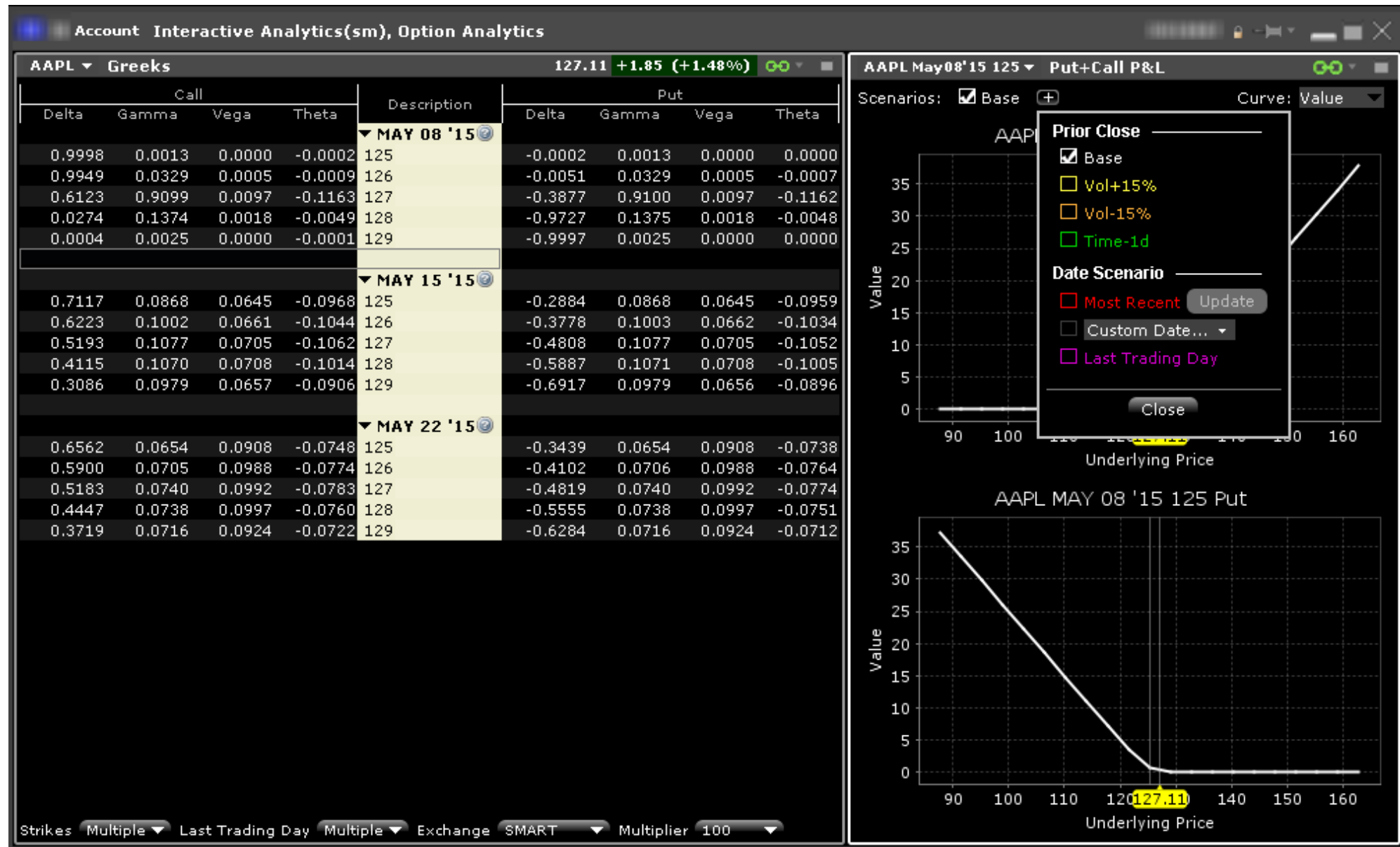
Vega

- The Vega ν of a portfolio is the rate of change of the value of the portfolio with respect to the volatility of the underlying asset
- If ν is highly positive or negative - the portfolio is very sensitive to small changes in volatility.
- If its close to zero, volatility changes have little impact on the value of the portfolio.

Rho

- The rho ρ of a portfolio is the rate of change of the value of the portfolio with respect to the interest rate
- Measures the sensitivity of the value of a portfolio to a change in the interest rate when all else is constant

Greeks interpretation



Delta close to 1 - the option is deep in the money - Gamma close to 0 - the option is deep in

the money

Scenario analysis

- We want to calculate the gain or loss on a portfolio over a specified time
 - With changing conditions

End

Dynamic aspects of delta hedging

- Consider $S_0 = \$49$, $K = \$50$, $r = 0.05$, $\sigma = 0.2$ and $T = 0.3846(20weeks)$
- Suppose the initial value of $\Delta = 0.522$ - then the Δ of the short position is -52,200
- As soon as the option is written 2, 557, 800 must be borrowed to buy 52,200 shares at \$49
- With $r = 5$ we have a cost of \$2, 500 in first week
- As stock price falls to \$48.12, the Δ falls to 0.458 (new Δ is -45,800).
- Sell 6,400 of the initial shares purchased to maintain the hedge.
- Strategy realizes 308, 00 in cash and cumulative borrowing in Week 1 reduced to 2, 252, 300
- Week 2, stock price is 47.37, Δ declines again, and so on.
- Towards the end of the option, option gets **exercised**
 - Δ approaches 1.0
 - In week 20, the hedger has a fully covered position (receive \$5million for the stock held)
 - Financial cost of **writing** the option and **hedging** it is \$263, 00

Week	Stock price	Delta	Shares purchased	Cost of shares purchased (\$000)	Cumulative cost including interest (\$000)	Interest cost (\$000)
0	49.00	0.522	52,200	2,557.8	2,557.8	2.5
1	48.12	0.458	(6,400)	(308.0)	2,252.3	2.2
2	47.37	0.400	(5,800)	(274.7)	1,979.8	1.9
3	50.25	0.596	19,600	984.9	2,966.6	2.9
4	51.75	0.693	9,700	502.0	3,471.5	3.3
5	53.12	0.774	8,100	430.3	3,905.1	3.8
6	53.00	0.771	(300)	(15.9)	3,893.0	3.7
7	51.87	0.706	(6,500)	(337.2)	3,559.5	3.4
8	51.38	0.674	(3,200)	(164.4)	3,398.5	3.3
9	53.00	0.787	11,300	598.9	4,000.7	3.8
10	49.88	0.550	(23,700)	(1,182.2)	2,822.3	2.7
11	48.50	0.413	(13,700)	(664.4)	2,160.6	2.1
12	49.88	0.542	12,900	643.5	2,806.2	2.7
13	50.37	0.591	4,900	246.8	3,055.7	2.9
14	52.13	0.768	17,700	922.7	3,981.3	3.8
15	51.88	0.759	(900)	(46.7)	3,938.4	3.8
16	52.87	0.865	10,600	560.4	4,502.6	4.3
17	54.87	0.978	11,300	620.0	5,126.9	4.9
18	54.62	0.990	1,200	65.5	5,197.3	5.0
19	55.87	1.000	1,000	55.9	5,258.2	5.1
20	57.25	1.000	0	0.0	5,263.3	

